Real Time Solar Forecasting And Intelligent EV Scheduling Using LSTM

Research Proposal and System Design Report

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Abstract

This project proposes an integrated energy management system for a commercial building equipped with solar photovoltaic panels and electric vehicle charging facilities. The system combines short-term solar forecasting with intelligent charging decisions so that available solar energy, electricity prices, building demand and vehicle departure requirements can be considered together. It extends two supplied studies: a comparison of machine learning models for solar forecasting and an LSTM-based method for commercial-building EV energy management [1, 2].

The proposed architecture uses one long short-term memory network to forecast photovoltaic output and a second LSTM to imitate charging decisions generated by an offline optimization model. A perfect-information mixed-integer linear program provides training targets and a cost benchmark. During operation, the learned scheduler uses only information available at the decision time. A separate constraint layer checks its actions, corrects infeasible allocations and records every intervention. Forecasts and schedules are refreshed at 15-minute intervals in the initial design.

The central experiment measures how forecast quality changes electricity cost, peak import, unmet charging demand and constraint interventions. Persistence, gradient boosting, immediate charging, a forecast-free learned policy and rolling-horizon optimization provide comparison baselines. Chronological evaluation, explicit dataset assumptions and repeated uncertainty experiments are used to make the comparison reproducible. A dashboard presents the operating state, with an optional language model summarizing verified metrics.

This report specifies a proposed implementation and evaluation method. No trained models, experimental outputs or field measurements were supplied, so expected benefits are presented as hypotheses rather than achieved results.

Keywords: solar forecasting, LSTM, electric vehicles, energy management, imitation learning, mixed-integer optimization, forecast uncertainty.

1 Introduction and Problem Statement

Background

Solar output varies with irradiance, clouds, temperature and the position of the sun. At the same time, commercial buildings experience changing electrical demand and vehicle arrivals. Charging every vehicle immediately can increase the grid import peak and leave midday solar generation underused. Deferring charging can reduce cost, but only when the remaining connection time and available charging capacity are sufficient to meet the user's energy requirement.

The solar forecasting study [1] demonstrates the usefulness of meteorological and solar geometry features. The EV management study [2] demonstrates an offline optimization and online LSTM approach to scheduling. Combining these ideas creates a decision problem: a forecast must be judged by the operating decisions it supports as well as by its numerical prediction error.

Problem statement

Design a system that repeatedly forecasts available solar power and allocates EV charging within site and vehicle limits, using only information available at that time. Determine whether adding explicit solar forecasts to an LSTM scheduler improves cost and charging service, and quantify how these outcomes change when forecasts become less accurate.

Research questions

- Does an LSTM forecast improve on persistence and gradient boosting under chronological testing?
- Does a forecast-informed LSTM scheduler offer a useful cost and computation trade-off relative to rolling-horizon optimization and a forecast-free policy?
- How do random error, persistent bias and timing errors affect cost, peak demand and departure energy shortfall?
- How often does the constraint layer alter the learned actions, and what operational trade-offs do those corrections create?

Objectives and scope

The project will develop a documented data pipeline, a forecasting model, an optimization benchmark, a learned scheduler and a reproducible evaluation. The minimum system uses 10-20 simulated charging ports driven by recorded session data, a configurable PV installation and a declared building-load scenario. Charging-only operation is the initial mode. Vehicle-to-grid discharge is an optional extension requiring explicit assumptions about bidirectional hardware, user consent and battery limits.

"Real time" refers to repeated supervisory decisions at a 15-minute cadence, with computation completed before the next control interval. Historical replay is the first validation setting. A functioning replay dashboard is not evidence of a live connection to chargers or of electrical grid stability.

2 Review of the Supplied Research Papers

Solar forecasting study

Babu et al. [1] compare linear, kernel, tree and ensemble regression models using atmospheric and solar geometric features. The abstract reports Gradient Boosting Regressor performance of $R^2 = 0.827$, RMSE = 399.44 and MAE = 253.62 on the paper's target scale. These are source-paper results, not results of the present project, and must not be treated as percentages or compared directly with a differently scaled PV installation.

The paper explicitly states in its model robustness discussion that validation used random sampling and identifies time-based and location-specific testing as future work [1, p. 93746]. This motivates chronological evaluation in the proposed project. It does not establish that every reported result is invalid or prove a particular leakage mechanism. The proposed study also adds a downstream EV scheduling evaluation, which is outside the paper's principal forecasting comparison.

LSTM energy management study

Zhou et al. [2] formulate a commercial-building energy management model and train an LSTM offline using optimization-derived information. Their online method maps available state information to scheduling outputs without an additional forecasting module. The study explicitly presents preliminary data processing and output filters; it would therefore be inaccurate to describe its filtering procedure as hidden.

The reported case contains 500 kW of onsite PV and 100 charging piles, with 250 weekdays divided into the first 220 days for training and the final 30 for validation. It uses 15-minute steps, a four-step rolling optimization horizon and an assumption of one EV per charging pile per weekday [2, pp. 1211-1212]. This assumption limits the variety of session turnover represented in that case.

The paper compares complete-information MILP, incomplete-information MILP, learned scheduling variants and a no-scheduling case. Its discussion uses a 90% prediction-accuracy assumption for the incomplete-information comparison [2, p. 1214]. The proposed project extends this treatment with precisely defined error distributions, multiple severity levels and a forecast-free ablation. It does not claim to be the first study to connect forecasting and control.

Combined research direction

The two papers motivate an integrated experimental system: retain meteorological and solar geometry features, learn temporal relationships with LSTM, use optimization as a benchmark, and evaluate the economic consequence of forecast error. The contribution sought here is a reproducible comparison with explicit information boundaries and constraint reporting. Whether this design outperforms established methods remains an empirical question.

3 Research Gaps and Proposed Responses

Observed scope or limitation	Proposed response	Evidence to collect
Random-sampling validation in the solar study [1]	Use chronological training, validation and held-out testing.	Errors by season, horizon and weather condition.
Forecasting and charging are assessed in separate supplied studies.	Evaluate forecasts through the same EV dispatch simulator.	Cost, import peak and unmet energy for each forecast.
One EV per pile per weekday in the EV case [2]	Replay multiple recorded sessions with port-availability checks.	Turnover, congestion and service completion.
A stated prediction-accuracy setting in the EV comparison [2]	Sweep defined error levels, bias and time displacement.	Measured forecast error against operating outcomes.
Output filtering is already present in [2].	Separate prediction from constraint enforcement and log corrections.	Intervention frequency, magnitude and residual violations.

Observed scope or limitation	Proposed response	Evidence to collect
The proposed system has not been experimentally evaluated.	Publish configurations, seeds, assumptions and negative findings.	Reproducible experiments and completed result tables.

Table 1. Research gaps identified in the supplied papers and the proposed responses.

Expected contribution

The project is intended to connect forecast quality to charging outcomes under a common dataset and simulator. A forecast-aware learned controller will be compared with a matched forecast-free controller, while a separately measured constraint layer will reveal how much of the final schedule comes from the model and how much comes from corrective control.

Claims that require evidence

Higher forecast accuracy does not automatically produce lower cost on every day. A cheaper schedule may also leave vehicles undercharged, so energy service must be reported alongside cost. Zero constraint interventions alone is not a sufficient quality measure: a policy that never charges may also require few corrections. Likewise, reduced peak import demonstrates load management but does not prove voltage, frequency or transient stability.

The project's distinction from the two supplied papers can be stated now. A claim of novelty across the wider literature would require a broader review and completed experiments. No publication acceptance, project grade or achieved performance is implied by this proposal.

4 Proposed System Architecture

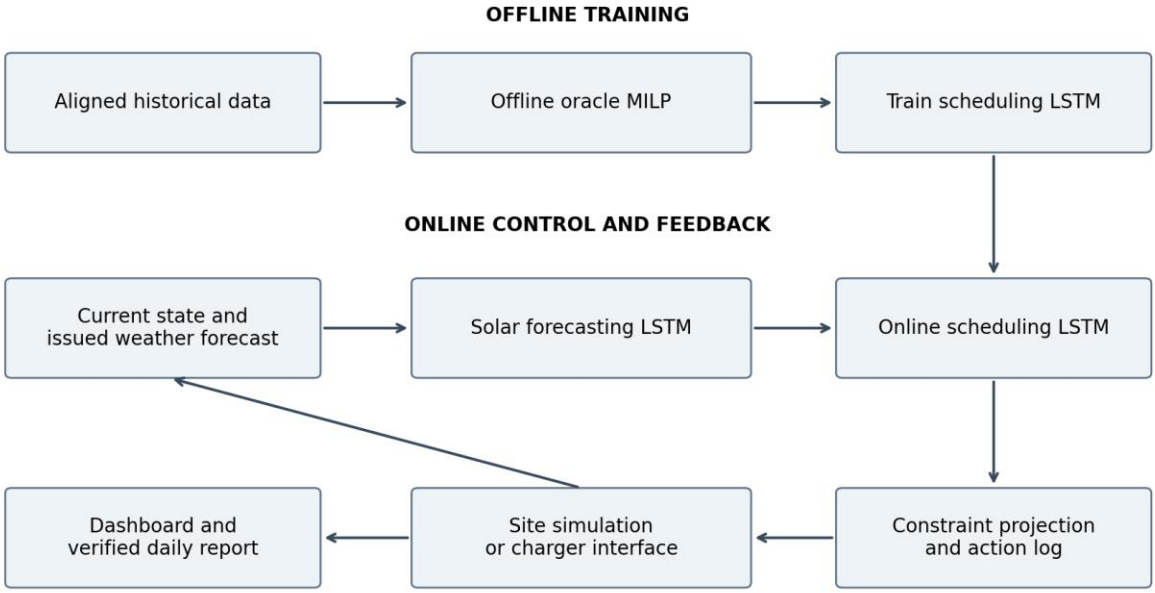


Figure 1. Offline optimization supplies learning targets. Online actions pass through constraint checks before execution. The feedback loop supplies the next measured state.

Offline workflow

Historical data are cleaned and divided chronologically. The solar forecaster is trained on the training period. Rolling out-of-fold forecasts are then generated within that period so that the scheduler sees realistic forecast errors rather than in-sample fitted values. The oracle solves historical episodes using complete future information to create action targets. Oracle outputs from held-out episodes are reserved for evaluation.

Online workflow

At each decision time, the system validates sensor freshness, updates connected sessions and generates a PV forecast over the remaining horizon. The scheduling LSTM proposes the next charging allocation. The constraint layer checks feasibility and modifies the allocation if necessary. Only the first interval is executed, after which measured energy and connection status update the state.

Information separation

The offline oracle may use future realized PV, load and departures. The online policy may use current measurements, past observations, issued forecasts, published tariffs and user declarations available at that instant. Actual future weather, final session energy and realized disconnect times must not enter its features. The report generator receives verified logs after control decisions; its text cannot change charging commands.

5 Data Requirements and Preparation

Data stream	Proposed source	Treatment
Weather and irradiance	Open-Meteo [4, 5]	Keep issue time, valid time, location and units.
EV charging sessions	Caltech ACN-Data [3]	Use connection events and available user requests.
Building demand	BDG2 or a declared scenario [6]	Document meter type, units and any profile scaling.
PV target	Measured PV where available	Otherwise identify irradiance-derived output as modeled PV.
Electricity tariff	Selected utility or synthetic tariff	Record currency, effective period and import/export prices.

Table 2. Proposed data streams, sources and preparation treatment.

Dataset compatibility

The initial study should select one EV site and obtain weather for its location. A building profile taken from another site or year creates a composite simulation; aligning timestamps does not turn it into a measured common-site dataset. BDG2 contains hourly records for 2016-2017 [6], so direct overlap with EV sessions and issued weather forecasts must be checked. Where dates do not overlap, document a scenario mapping instead of claiming joint historical observation.

Temporal and energy consistency

Store timestamps in UTC and preserve local time for tariffs and daily features. Use 15-minute control intervals. Convert interval energy to average power using $P = E / \Delta t$, with $\Delta t = 0.25$ h. For hourly building energy distributed over four intervals, divide the hourly kWh by four; this preserves energy but does not reveal real sub-hour variability. Check missing values, duplicates, daylight-saving transitions, impossible session durations and overlapping use of the same port.

What the controller can know

ACN-Data distinguishes delivered energy from optional user inputs, including requested energy [3]. Use only declarations available at the scheduling time. When a declaration is absent, estimate the requirement and departure from prior training data, or state a simulated request. Battery capacity, initial SOC and discharge permission require separate assumptions unless independently measured. Delivered energy is not automatically the driver's requested energy.

Open-Meteo historical weather is reanalysis, whereas historical forecast products serve different purposes [4, 5]. A deployment-like backtest must retrieve forecasts available before the decision time and preserve their lead times. If that archive is unavailable for the study dates, use a causal history-only forecaster and describe its limitation. Future realized weather is permitted only in an explicitly labeled oracle experiment.

Data split

Start with a chronological 60% training, 20% validation and 20% test split, adjusted if data coverage is insufficient. Fit scaling, missing-value rules and feature selection only on training data. Keep target windows inside their respective periods, prevent a session from leaking across partitions, and report the exact dates and retained sample counts.

6 Solar Forecasting Using LSTM

Inputs and prediction horizon

The forecasting model receives recent PV or irradiance history, temperature, cloud information when available, lagged building-independent weather features and deterministic time features. Solar zenith and azimuth are computed for the configured location and timestamp, following the feature direction of [1]. Weather forecasts must carry their original issue times. The first model uses the previous 24 hours of observations and predicts up to 96 future 15-minute intervals, refreshed at every control step.

Model design

A starting architecture uses two LSTM layers with 64 and 32 hidden units, a dropout rate of 0.2 and a dense output layer for the forecast horizon. These values are proposed configuration choices, not tuned results. The LSTM cell stores a recurrent memory state and uses input, forget and output gates to decide which temporal information to retain.

Training uses normalized inputs, a mean squared error objective, early stopping and validation-selected checkpoints.

Predicted power is restricted to physically plausible values between zero and the declared PV capacity, with nighttime power set to zero using solar geometry. Report metrics after the same physical postprocessing for every competing model. If the target is calculated from irradiance instead of measured at an inverter, state that the experiment evaluates modeled generation. A forecast of modeled PV is weaker evidence for performance on a real installation.

Baselines and uncertainty

Compare the LSTM with same-time previous-day persistence and gradient boosting using equivalent available features. Evaluate a short-horizon persistence variant separately when appropriate. Keep LSTM in the proposed architecture, but retain the best validated model as a deployment candidate if a simpler baseline performs better.

For uncertainty, estimate prediction intervals from a held-out chronological calibration block or train quantile outputs. A proposed 90% prediction interval must be checked for empirical coverage and average width by lead time and daylight condition. It is not a guarantee that every future observation has 90% coverage. For the first scheduler, use the point forecast; uncertainty-aware scheduling is a separately identified extension.

Forecast evaluation

Report MAE and RMSE in kW, normalized RMSE as a percentage of installed PV capacity, and forecast skill relative to persistence: skill = 1 − RMSE(model) / RMSE(persistence). Omit the skill ratio when its denominator is zero. Report both all-hour and daylight-only scores because nighttime zeros can make aggregate errors look deceptively small. Separate one-hour, several-hour and day-ahead horizons.

Inspect sunny days, cloudy days and rapid output ramps separately. Save residuals by issue time and horizon so the subsequent scheduling experiment can use empirical error patterns. Forecast selection uses the validation set; the final test period remains untouched until the experimental configuration is fixed.

7 Optimization Model and Oracle Benchmark

The optimization model schedules EV charging to minimize electricity cost while meeting vehicle energy requirements and site limits. Power is measured in kW, energy in kWh, and the scheduling interval is Δt.

Objective and Constraints

The objective includes electricity import cost, export revenue, optional battery degradation cost and a penalty for unmet charging demand:

$$J = \Sigma t\, \Delta t\, [\, ctb\, g{+}t - cts\, g{-}t + cdeg\, \Sigma i\, (pci,t + pdi,t)\,] + \lambda\, \Sigma i\, si$$

Here, g+ and g− represent grid import and export, pc and pd represent charging and discharging power, and si is departure energy shortfall. Preferably, unmet demand is minimized first, followed by operating cost.

The model enforces:

- Balance between grid power, usable solar generation, building demand and EV power.
- Battery energy limits and departure requirements.
- Charging only while vehicles are connected.
- Charger and grid power limits.
- No simultaneous charging/discharging or grid import/export.

Discharging is disabled in the initial charging-only implementation.

Oracle and Rolling Horizon Comparison

The oracle uses complete future information to establish the minimum achievable cost under identical assumptions. This benchmark requires a certified optimal solution; solver status and optimality gap must be reported.

Rolling-horizon optimization uses available forecasts, executes the next interval and updates its schedule. It receives the same information as the learned controller for a fair comparison.

Monthly demand charges are initially excluded. Charging commitments must continue across horizon boundaries so that postponing unmet demand does not create artificial savings.

8 Learned Scheduling and Constraint Enforcement

Training the Scheduling LSTM

The second LSTM learns charging decisions from the offline oracle. Inputs include solar forecasts, building demand, tariffs, vehicle energy requirements and departure times. It outputs charging power for each connected port.

Training uses oracle actions and forecasts generated without access to future observations. Performance is evaluated over complete charging episodes to capture the effects of earlier decisions. Any corrective retraining uses training data only.

Feasibility Projection

A separate constraint layer checks and adjusts the predicted charging allocation. It disables disconnected ports, enforces charger limits and jointly allocates available site capacity using current solar generation and building demand.

The layer also checks remaining energy needs against time to departure. Since vehicles share limited capacity, checking only the next interval cannot guarantee that all departure requirements will be met.

Fallback and Audit Trail

If the model or constraint correction fails, the system prioritizes vehicles with earlier departures while respecting available capacity and reporting unmet demand. Building overloads, unexpected disconnections and outdated measurements are logged separately.

Logs record proposed and executed power, corrections, violated constraints and failure reasons. Evaluation reports intervention frequency and remaining violations. Compliance depends on the modeled assumptions; hardware protection remains separate from supervisory scheduling.

9 Experimental Design and Evaluation

Comparison Methods

Method	Information available	Purpose
Immediate charging	Current requests and site capacity	Basic charging reference
Rule-based allocation	Current state, tariff and departure times	Practical comparator
Rolling-horizon optimization	Forecasts and current state	Optimization baseline
Forecast-free LSTM	Current and past state	Test performance without solar forecasts
Forecast-informed LSTM	Same inputs plus PV forecasts	Evaluate the proposed approach
Oracle optimization	Complete future information	Offline benchmark

Table 3. Comparison methods used in the experimental design.

Controlled Forecast Sensitivity

Test the fixed trained scheduler at target forecast-error levels of 0%, 5%, 10% and 20%, using:

$$nRMSE = (RMSE / Installed\ PV\ Capacity) \times 100$$

Create controlled forecasts by adding noise, bias or timing shifts to actual PV output. Restrict predictions to physical limits and set nighttime generation to zero. Report achieved error after these adjustments. The 0% case provides perfect solar knowledge only. Evaluate actual model forecasts separately from these synthetic tests.

Fairness and Repetition

Compare methods using identical EV sessions, tariffs, site limits and actual solar generation. Apply matching forecast errors, use at least five noise seeds and keep policies unchanged during testing.

Measure operating cost, unmet charging demand, peak grid import and constraint interventions across different weather and occupancy conditions. Report averages and uncertainty across days or complete episodes. Include timeouts, infeasible schedules and unmet requests in the results rather than discarding them.

10 Performance Measures and Result Reporting

Performance Measures

Measure	Evaluation
Forecast error	MAE, RMSE and capacity-normalized RMSE for daylight and all hours
Operating cost	Import cost minus export revenue, plus battery throughput cost
Savings	Reference cost − method cost; percentage only for a positive reference cost
Oracle cost gap	Method cost − certified oracle optimum under identical conditions
Peak import	Maximum grid power in kW over the same observation interval
Charging service	Unmet energy, fully served sessions and early-disconnect failures

Measure	Evaluation
Constraint performance	Intervention frequency, correction magnitude and remaining violations
Computation	Median and 95th-percentile decision time, hardware and timeouts

Table 4. Performance measures and evaluation basis.

Results Presentation

Present data coverage, forecast accuracy by horizon and scheduling outcomes on identical test episodes. Plot operating cost and unmet energy against achieved forecast error. Evaluate the constraint layer's effect through before-and-after comparisons in simulation.

All experimental results are not yet measured. Validation requires held-out forecasts, paired scheduling trials, repeated error experiments, action logs and execution timings.

Expected Outcomes

Accurate solar forecasts may improve charging decisions when schedules are flexible and electricity prices vary. The learned scheduler may reduce computation, but lower costs and faster execution must be demonstrated through fair comparisons.

Illustrative savings are not experimental results. Actual performance depends on tariffs, PV capacity, demand and charging requirements. A simpler baseline outperforming the LSTM remains a valid research finding.

11 Implementation and Development Plan

Software Modules

The system includes data preparation, solar forecasting, optimization, LSTM scheduling, constraint enforcement and evaluation. Python supports development, Pyomo and HiGHS handle optimization, FastAPI serves forecasts and schedules, and Streamlit provides the dashboard. Container packaging enables repeatable demonstrations.

Store dataset details, time-based splits, model checkpoints, random seeds and solver settings for reproducibility. Log forecasts and charging actions with timestamps, units and feasibility status.

Dashboard and Reporting

The dashboard displays solar forecasts, building demand, grid import, EV schedules, remaining energy needs, costs and constraint interventions. Clearly distinguish simulation, historical replay and live operation, and allow results to be exported.

An optional language model summarizes verified metrics and logged decisions. Its numerical statements must be checked, and it cannot issue charging commands.

Development Schedule

Period	Main tasks	Deliverable
Weeks 1-2	Clean and align datasets	Documented dataset and assumptions
Weeks 3-4	Develop forecasting baselines and LSTM	Validation report
Weeks 5-6	Build optimizer and constraint layer	Checked benchmark schedules
Weeks 7-8	Train and evaluate scheduling LSTM	Comparison with baselines
Weeks 9-10	Conduct sensitivity tests and ablations	Results tables and figures
Weeks 11-12	Complete dashboard and report	Packaged demonstration and findings

Table 5. Proposed development schedule.

Constraint enforcement must precede scheduling demonstrations. If time is limited, reduce the number of ports or simplify building demand, and defer optional features while retaining fair evaluation and forecast-sensitivity testing.

12 Limitations and Conclusion

Remaining Limitations

Combining datasets from different sites creates a simulated scenario requiring real-world validation. Estimated PV generation, missing vehicle requests and assumed battery properties introduce uncertainty.

The learned scheduler may perform differently under new tariffs, weather or charging conditions. Constraint enforcement depends on accurate measurements and cannot satisfy infeasible energy requests. Simulation also cannot capture every communication delay or equipment fault. Future work includes joint uncertainty modeling, hardware

testing and electrical network analysis before claiming grid stability improvements.

Conclusion

Real Time Solar Forecasting And Intelligent EV Scheduling Using LSTM integrates solar prediction with optimization-trained EV scheduling. It extends the two base papers through chronological evaluation, controlled forecast-error experiments and transparent constraint reporting.

The proposed system will be evaluated for operating cost, charging service, computational efficiency and constraint compliance. Performance conclusions will follow completed experiments rather than assumed improvements.

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Source and Evidence Note

References [1]-[2] provide the research foundation, [3]-[6] describe supporting data resources, and [7] outlines the proposed solution. System settings and expected outcomes remain proposals; no completed experiments or measured results are claimed.

Planned Deliverables

- A reproducible data pipeline with documented assumptions and time-based splits.
- A trained solar forecasting LSTM and baseline comparison.
- Oracle and rolling-horizon optimizers, plus a learned scheduler with constraint enforcement.
- Forecast-sensitivity plots, service metrics, action logs and execution timings.
- An interactive demonstration and experimental report with references.